Executive Strategy Memo

Optimizing Delivery ETAs with Graph-Based Network Intelligence

The Information Gap

Delhivery's route planning currently relies on standard OSRM estimates that assume frictionless roads and zero structural congestion. In reality, shipments stall, SLAs are missed, and costs cascade silently through the network—not by chance, but through mathematically identifiable bottlenecks that standard tools cannot see. By modeling our network as a directed weighted graph across 144,867 logged trips, we have pinpointed the exact facilities bleeding time and money. Here are the Top 5 network offenders, ranked by their structural influence, chronic lane failure count, and peak delay amplification.

Top 5 Network Offenders

1. GURGAON – BILASPUR HUB *Haryana · IND000000ACB*

SLA IMPACT SCORE	CHRONIC CORRIDORS	PEAK DELAY FACTOR
100.0 / 100	85	5.45x

Highest network centrality of any hub.
Disruption here propagates the farthest and fastest across the entire logistics system. It is the singular highest-priority intervention target for immediate capacity scaling and routing bypass.

2. BANGALORE – NELAMANGALA HUB *Karnataka · IND562132AAA*

SLA IMPACT SCORE	CHRONIC CORRIDORS	PEAK DELAY FACTOR
55.1 / 100	68	4.00x

Second-highest chronic corridor burden.
As South India's primary e-commerce gateway, persistent underperformance here carries outsized commercial consequences. Requires critical structural upgrades and FTL vs. Carting framework implementation.

3. KOLKATA – DANKUNI HUB *West Bengal · IND712311AAA*

43.3 / 100	46	9.94x
SLA IMPACT SCORE	CHRONIC CORRIDORS	PEAK DELAY FACTOR

The most severe delay amplifier.
Shipments regularly take nearly 10x longer than estimated—a systemic SLA liability for all East India flows. The extreme peak delay indicates total facility saturation during peak hours.

4. BHIWANDI – MANKOLI HUB *Maharashtra · IND421302AAG*

12.7 / 100	58	5.57x
SLA IMPACT SCORE	CHRONIC CORRIDORS	PEAK DELAY FACTOR

Severe Western region chokepoint.
A high volume of chronic lanes combined with significant delay multipliers creates intense localized capacity strain. FTL offloading and intelligent shipment rerouting are highly advised.

5. HYDERABAD – SHAMSHABAD HUB *Telangana · IND501359AAE*

11.1 / 100	56	9.94x
SLA IMPACT SCORE	CHRONIC CORRIDORS	PEAK DELAY FACTOR

Extreme secondary delay amplifier.
Matches Kolkata with a near 10x delay magnification. This excessive peak delay factor points to severe structural congestion and dwell-time breakdowns during peak operational windows.